From Wealth Inequality to Price Dispersion of Cosmetics: An Econophysics Approach

Ying-Shan CHOU[1], Chi-Ning CHEN[2], and Chung-I CHOU[3]

[1,2] Department of Physics, National Dong-Hwa University, Hualien 974, Taiwan

[3] Department of Physics, Chinese Culture University, Taipei, 111, Taiwan

E-mail: [3]cichou@faculty.pccu.edu.tw

Abstract

We analyzed tens of thousands of cosmetic products from the cosmetic databases of Taiwan and Japan on Internet. The results show the long tail (or power law decay) behavior in the cumulative distribution function (CDF) of prices of cosmetics in Taiwan and Japan.

We attribute such behavior to the purchasing power (or wealth) of consumers. To confirm this assumption, we have studied the wealth distribution of Taiwan, and also carried out a questionnaire survey from one thousand university students and staffs. Our results show that there is in fact correlation between wealth and consumption expenditure.

To understand the above results, we constructed a simple pricing model based on the agent-based approach. The basic ingredients of this model are the wealth distribution, buying strategy of consumers, and pricing strategy of business owners. The results of simulation of our model show: i) power-law distributions can indeed be produced in the simulation of the model; ii) the price distributions of cosmetics do depend on the wealth distributions.

Keyword: long tail behavior, wealth distributions, agent-based model